

GALOIS COHOMOLOGY OF ELLIPTIC CURVES OVER THE ANTICYCLOTOMIC EXTENSION

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1. INTRODUCTION

Let K be an imaginary quadratic number field and $p \geq 3$ be an odd prime number. By Leopoldt's conjecture, which is known to hold over K , there are two linearly independent $\mathbb{Z}_p$ -extensions, namely the cyclotomic $\mathbb{Z}_p$ -extension and the anticyclotomic $\mathbb{Z}_p$ -extension, which we denote by K_{cyc} and K_{ac} respectively. Let E be an elliptic curve over K which is ordinary at the primes above p . Iwasawa theory of elliptic curves over cyclotomic $\mathbb{Z}_p$ -extensions has been studied for almost five decades, and that over the anticyclotomic extension spans a period of smaller duration. The results are significantly different over these two extensions. While a conjecture of Mazur, later proved by Kato, asserts that the dual Selmer group $X(E/K_{\text{cyc}})$ over the cyclotomic extension is torsion as a module over the corresponding Iwasawa algebra, the dual Selmer group $X(E/K_{\text{ac}})$ is not necessarily torsion as a module over the corresponding Iwasawa algebra and may have positive rank. When the rank is positive, the contribution to the rank is related to the existence of Heegner points in the Mordell-Weil group of the elliptic curve along the anticyclotomic extension. Iwasawa theory of E in the anticyclotomic setting has been studied by Perrin-Riou [PR84, PR87, PR92, PR95] and later by Howard [How04a, How04b], Bertolini-Darmon [BD05], Agboola-Howard [AH06] depending on the different contexts that are possible in this setting.

The goal in this article is to approach the Iwasawa theoretic study of elliptic curves over anticyclotomic $\mathbb{Z}_p$ -extensions from within a unified framework that is largely algebraic. In the case that the dual Selmer group is torsion over the anticyclotomic extension, we show that the methods and results are analogous to those in the cyclotomic extension. If the dual Selmer group over the anticyclotomic extension has positive rank, then we recover most of the known results by far simpler methods. Our approach and techniques have the advantage that they can be used to illustrate the main results with concrete numerical examples.

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This article consists of four sections, including this introductory section. Section 2 is preliminary in nature. In Section 3, we recall results in the literature regarding the module structure of the Selmer group over the anticyclotomic extension. In both Section 4.1 and Section 4.2, we study the Galois cohomology of Selmer groups in the generic case. In Section 4.1, we study the module structure of the Selmer group over the compositum $K_{\text{cyc}} \cdot K_{\text{ac}}$. In Section 4.2, we recover the results in [Ber01] with different but simpler hypotheses.

2. PRELIMINARIES

2.1. Selmer groups. Let p be an odd prime. Let K be a number field and E/K be an elliptic curve which is ordinary at every prime above p . Denote by S a finite set of primes containing the primes above p and the primes of bad reduction of E/K . Let K_S be the maximal extension that is unramified outside of S .

For every Galois extension L/K contained in K_S and a prime v in K , we will define the group $J_v(E/L)$ as

$$J_v(E/L) = \bigoplus_{w|v} H^1(L_w, E)(p)$$

where the direct sum is over the primes w of L lying above v . For a finite extension L/K , we define the p^n -Selmer group $\text{Sel}(E_{p^n}/L)$ as the kernel of the natural restriction map

$$H^1(K_S/L, E_{p^n}) \rightarrow \bigoplus_{v \in S} J_v(E/L),$$

and define the p -primary Selmer group $\text{Sel}_p(E/L)$ as the direct limit

$$\text{Sel}_p(E/L) := \varinjlim_n \text{Sel}(E_{p^n}/L).$$

For a general Galois extension L/K , we define $\text{Sel}_p(E/L) = \varinjlim_F \text{Sel}_p(E/F)$ where F runs over the finite extensions of K contained in L . If L is contained in K_S , then there is an exact sequence

$$0 \rightarrow \text{Sel}_p(E/L) \rightarrow H^1(K_S/L, E_{p^\infty}) \rightarrow \bigoplus_{v \in S} J_v(E/L).$$

For a finite Galois extension L/K , we define the compact Selmer group as the projective limit $\mathcal{S}_p(E/L) := \varprojlim_n \text{Sel}_p(E_{p^n}/F)$ with respect to the projections $E_{p^{n+1}} \xrightarrow{\times p} E_{p^n}$. For a general Galois extension L/K , let $\mathcal{S}_p(E/L)$ be the inverse limit $\mathcal{S}_p(E/L) := \varprojlim_F \mathcal{S}_p(E/F)$ with respect to corestriction maps, where F runs over the finite extensions of K contained in L .

2.2. Modules over the Iwasawa algebra. Let $\mathcal{L}/K$ be a p -adic Lie extension. For a $\mathbb{Z}_p$ -module A , we will denote by

$$A^\vee = \text{Hom}(A, \mathbb{Q}_p/\mathbb{Z}_p)$$

its Pontryagin dual. The p -primary group $\text{Sel}_p(E/\mathcal{L})$ has the discrete topology as a $\Lambda(G)$ -module and its Pontryagin dual $X(E/\mathcal{L}) := \text{Hom}(\text{Sel}_p(E/\mathcal{L}), \mathbb{Q}_p/\mathbb{Z}_p)$ is a compact $\Lambda(G)$ -module.

The natural action of $G = \text{Gal}(\mathcal{L}/K)$ on $\text{Sel}_p(E/\mathcal{L})$ is continuous, and these groups are naturally considered as modules over the Iwasawa algebra [Gre99, Section 1]

$$\mathbb{Z}_p[[\text{Gal}(K_\infty/K)]] := \varprojlim_{F \subset \mathcal{L}, F/K \text{ finite}} \mathbb{Z}_p[\text{Gal}(F/K)].$$

We will denote by $\Lambda(G)$ the Iwasawa algebra attached to the group G . When $G \simeq \mathbb{Z}_p$ and X is finitely-generated over $\Lambda(G)$, we will denote the Iwasawa μ and λ -invariants of X by $\mu_G(X)$ and $\lambda_G(X)$ respectively. The group G is often clear from the context, in which case we will suppress G from the notation and let $\mu(X) := \mu_G(X)$, $\lambda(X) := \lambda_G(X)$.

3. MODULE STRUCTURE OF SELMER GROUPS OVER THE ANTICYCLOTOMIC EXTENSION

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N , $p \geq 5$ be a prime where E has good ordinary reduction. Let K be an imaginary quadratic field of discriminant $\text{disc}(K) \neq -3, -4$. We may decompose $N = N^+ N^-$ where N^+ is the product of the primes dividing N that split in $K/\mathbb{Q}$ and N^- is the product of the primes that are inert. Following the terminology in [Ber01], we say that E is in the generic case if E does not have complex multiplication (CM) or the field of CM of E is different from K . Otherwise, we say that E is in the exceptional case.

Denote by K_{cyc} and K_{ac} the cyclotomic and anticyclotomic extensions over K , respectively. Moreover, let $K_\infty = K_{\text{ac}} \cdot K_{\text{cyc}}$ be their compositum and let $G_\infty = \text{Gal}(K_\infty/K)$. The Galois groups $\Gamma_{\text{cyc}}, H_{\text{cyc}}, \Gamma_{\text{ac}}, H_{\text{ac}}$ are defined as follows:

$$\begin{aligned} \Gamma_{\text{cyc}} &= \text{Gal}(K_{\text{cyc}}/K), H_{\text{cyc}} = \text{Gal}(K_\infty/K_{\text{cyc}}), \\ \Gamma_{\text{ac}} &= \text{Gal}(K_{\text{ac}}/K), H_{\text{ac}} = \text{Gal}(K_\infty/K_{\text{ac}}). \end{aligned}$$

3.1. Generic case. First consider the case when the complex L -function $L(E/K, s)$ vanishes to even order at $s = 1$. This occurs, for example, when N^- is a square-free product of an odd number of primes. In this case, it is due to Bertolini-Darmon that $X(E/K_{\text{ac}})$ is $\Lambda(\Gamma_{\text{ac}})$ -torsion under mild technical hypotheses [BD05]. This is also known as the

definite setting in the literature. On the other hand, in the case where N^- is an even product of number of primes, the module $X(E/K_{\text{ac}})$ has $\Lambda(\Gamma_{\text{ac}})$ -rank 1 [How04a, How04b]. This is known as the *indefinite* setting. The reasoning behind such terminology is related to the quaternion algebra over $\mathbb{Q}$ whose discriminant has finite part equal to N^- and the study of its associated Shimura variety. For further details on this terminology, readers are advised to consult [BLV23, Section 2].

Let us first consider the definite setting. Because $X(E/K_{\text{ac}})$ is $\Lambda(\Gamma_{\text{ac}})$ -torsion, the result of Greenberg [Gre89, Section 7] allows us to conclude that $X(E/K_{\text{ac}})$ does not have any non-trivial finite $\Lambda(\Gamma_{\text{ac}})$ -submodules. Greenberg's result concerns Selmer groups of general p -adic Galois representations and the statement below is a simplified version for elliptic curves due to Bertolini.

Theorem 3.1. [Ber01, Theorem 4.5] *Let $\mathcal{T}$ be the order of the torsion group of $\bigoplus_{v \in S \setminus \{p\}} E(K_v)$. Assume that*

- (1) $p \nmid 6\mathcal{T}$,
- (2) $X(E/K_{\text{ac}})$ is a $\Lambda(\Gamma_{\text{ac}})$ -torsion module.

Then $X(E/K_{\text{ac}})$ has no non-trivial finite $\Lambda(\Gamma_{\text{ac}})$ -submodules.

It should be noted that this statement applies more generally to any $\mathbb{Z}_p$ -extension, and the key hypothesis is that $X(E/K_{\text{ac}})$ is a $\Lambda(\Gamma_{\text{ac}})$ -torsion module.

When $E(K)$ is finite, one can readily deduce that $X(E/K_{\text{ac}})$ is $\Lambda(\Gamma_{\text{ac}})$ -torsion, by using the well-known fact that the control morphism $\text{Sel}_p(E/K) \rightarrow \text{Sel}_p(E/K_{\text{ac}})^{\Gamma_{\text{ac}}}$ is always a pseudo-isomorphism. In this case, the argument of Theorem 3.11 in the work of Coates and the second author [CS00] in the cyclotomic setting can be applied to give an alternative proof of the non-existence of non-trivial finite $\Lambda(\Gamma_{\text{ac}})$ -submodules.

Proposition 3.2. [Ven02, Proposition 3.10] *Let G be a p -adic Lie group. Then a compact $\Lambda(G)$ -module M does not have any non-trivial pseudo-null $\Lambda(G)$ -submodules if $\text{pd}_{\Lambda(G)}(M) \leq 1$.*

Proposition 3.3. *Let $\Gamma \simeq \mathbb{Z}_p$ and M be a compact $\Lambda(\Gamma)$ -module such that $H_1(\Gamma, M) = 0$. Then M does not have any non-trivial finite submodules.*

Proof. Let $\text{pd}_{\Lambda(\Gamma)}(M)$ and $\text{depth}(M)$ respectively denote the projective dimension and depth of M as a $\Lambda(\Gamma)$ -module.

When $H_1(\Gamma, M) = 0$ we have $\text{depth}(M) \geq 1$, which means $\text{pd}_{\Lambda(G)} M \leq 1$ using the Auslander-Buchsbaum formula [Eis95, Theorem 19.9]

$$\text{pd}_{\Lambda(\Gamma)}(M) + \text{depth}(M) = \dim(\Lambda(\Gamma)) = 2.$$

The result now follows from Proposition 3.2. An alternative proof is also given in [CS00, Lemma A.1.5]. $\square$

Using Proposition Proposition 3.3, we obtain

Theorem 3.4. [CS00, Theorem 3.11] *Assume that*

- (1) $p > 2$,
- (2) $\text{Sel}_p(E/K)$ is finite,
- (3) For each $P \neq 0$ in $E(K)(p)$, there exists a place v in K dividing p such that the reduction of P modulo v is nonzero.

Then $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = 0$, and the module $X(E/K_{ac})$ does not have any non-trivial finite $\Lambda(\Gamma_{ac})$ -submodules.

The argument of Theorem 3.1 no longer holds in the indefinite setting because $X(E/K_{ac})$ is not $\Lambda(\Gamma_{ac})$ -torsion. Nonetheless, Bertolini was able to show that $X(E/K_{ac})$ does not have any non-trivial finite $\Lambda(\Gamma_{ac})$ -submodules under certain hypotheses.

For a prime v in K above p , denote by $\tilde{E}_v$ the reduction of E/K modulo v and let k_v be the residue field of K_v . Moreover, let $d_v = \#\tilde{E}_v(k_v)$. For each prime $\ell \mid N$, denote by $c_\ell(E/\mathbb{Q})$ the Tamagawa factor of $E/\mathbb{Q}$ at ℓ . For an integer n , let $n^{(p)}$ be the largest prime power that divides n .

Theorem 3.5. [Ber01, Theorem 7.1] *Assume that*

- (1) $p \nmid 6N \text{disc}(K) \# \text{Pic}(\mathcal{O}_K)$,
- (2) $\bar{\rho}_{E,p} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_p)$ is surjective,
- (3) $p \nmid \prod_{v|p} \#\tilde{E}_v(k_v) \cdot \prod_{\ell \mid N} c_\ell(E/\mathbb{Q})$,
- (4) The module of Heegner points over K_n is non-zero modulo $pE(K_n)$ for some $n \geq 0$,
- (5) The natural map $\text{III}(E/K_n) \rightarrow \text{III}(E/K_{n+1})$ is an injection for all $n \geq 0$.

Then $X(E/K_{ac})$ does not have any non-trivial finite $\Lambda(\Gamma_{ac})$ -submodule.

Remark 3.6. In Section 4.1, we shall use Proposition 3.2 and Proposition 3.3 to prove the non-existence of non-trivial pseudo-null submodules of $X(E/K_\infty)$. In Section 4.2, we shall use Proposition 3.3 to prove the non-existence of non-trivial finite submodules of $X(E/K_{ac})$. Our arguments vastly simplify the proofs of such results in the existing literature.

3.2. Exceptional case. We recall that in this case, $E/\mathbb{Q}$ is an elliptic curve with CM by $\mathcal{O}_K$. We also note that in this case, K must be an imaginary quadratic field of class number 1 by CM theory. Moreover, $K_\infty \subset K(E_{p^\infty})$ and there is an isomorphism $\text{Gal}(K(E_{p^\infty})/K) \simeq \text{Gal}(K_\infty/K) \times \text{Gal}(K(E_p)/K)$. It is a result due to Coates [Coe83] that the dual Selmer group $X(E/K_\infty)$ is always $\Lambda(G_\infty)$ -torsion. When the sign of the functional equation of $L(E/\mathbb{Q}, s)$ is $+1$, the module $X(E/K_{ac})$ is $\Lambda(\Gamma_{ac})$ -torsion [Gre83, Theorem 3]. Otherwise, when the sign of the functional equation of $L(E/\mathbb{Q}, s)$ is -1 , the module $X(E/K_{ac})$ has $\Lambda(\Gamma_{ac})$ -rank 2 [AH06, Theorem A]. In fact, it was known to Greenberg that $X(E/K_{ac})$ is not $\Lambda(\Gamma_{ac})$ -torsion when the sign is -1 [Gre83, Theorem 2].

Now, let M_∞ be the maximal pro- p abelian extension over K_∞ that is unramified outside p . The Galois group $\text{Gal}(M_\infty/K_\infty)$ has a natural $\Lambda(G_\infty)$ -module structure and was studied by Greenberg [Gre78]. The module structures of $X(E/K_\infty)$ and $\text{Gal}(M_\infty/K_\infty)$ are related by the following result of Perrin-Riou:

Theorem 3.7. [PR84, Théorème 23] *Suppose that $X(E/K_\infty)$ is $\Lambda(G_\infty)$ -torsion. Then $X(E/K_\infty)$ does not have any non-trivial finite $\Lambda(G_\infty)$ -submodules if and only if $\text{Gal}(M_\infty/K_\infty)$ does not have any non-trivial finite $\Lambda(G_\infty)$ -submodules.*

Greenberg showed that $\text{Gal}(M_\infty/K_\infty)$ contains no non-trivial pseudo-null submodules [Gre78], from which it follows that

Theorem 3.8. *The $\Lambda(G_\infty)$ -module $X(E/K_\infty)$ does not have any non-trivial pseudo-null $\Lambda(G_\infty)$ -submodules.*

For each $\alpha \in \mathcal{O}_K$, let E_α denote the kernel of α as a K -endomorphism of E . For an ideal $\mathfrak{a} \subset \mathcal{O}_K$, denote by $E_{\mathfrak{a}}$ the $\mathfrak{a}$ -torsion points of E , i.e. $E_{\mathfrak{a}} = \bigcap_{\alpha \in \mathfrak{a}} E_\alpha$. Now, denote by $E_{p^\infty} = \bigcup_{n \geq 0} E_{p^n}$.

Assuming the Leopoldt conjecture holds for $K(E_p)$ and the weak Leopoldt conjecture holds for $K(E_{p^\infty})/K(E_p)$ [PR84, p. 44], we have the following result.

Theorem 3.9. [PR84, Théorème 25] *The $\Lambda(\Gamma_{ac})$ -module $X(E/K_{ac})$ does not have any non-trivial finite $\Lambda(\Gamma_{ac})$ -submodules.*

Proof. Under the condition that the Leopoldt conjecture holds for $K(E_p)$ and the weak Leopoldt conjecture holds for $K(E_{p^\infty})/K(E_p)$, Perrin-Riou showed $\text{pd}_{\Lambda(G_\infty)}(X(E/K_\infty)) \leq 1$.

Moreover, there is an isomorphism $X(E/K_{ac}) \simeq X(E/K_\infty)_{H_{ac}}$ by [PR84, Lemme 9, Proposition 12]. Therefore,

$$\text{pd}_{\Lambda(\Gamma_{ac})}(X(E/K_{ac})) = \text{pd}_{\Lambda(\Gamma_{ac})}(X(E/K_\infty)_{H_{ac}}) = \text{pd}_{\Lambda(G_\infty)}(X(E/K_\infty)) \leq 1$$

by Lemme 5 in [PR84]. The conclusion now follows from another application of Proposition 3.2. $\square$

4. GALOIS COHOMOLOGY OF THE SELMER GROUP

4.1. The definite setting. Recall that in this setting, N^- is a square-free product of an odd number of primes [BD05]. We work under the extra hypothesis that $\text{Sel}_p(E/K)$ is finite.

Hypothesis 4.1. Assume that the following hold:

- (surj) The residual Galois representation $\bar{\rho}_{E,p} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_p)$ is surjective.
- (finite) The Selmer group $\text{Sel}_p(E/K)$ is finite.

Hypothesis (finite) implies that $X(E/K_{ac})$ is $\Lambda(\Gamma_{ac})$ -torsion, because the restriction map $\text{Sel}_p(E/K) \rightarrow \text{Sel}_p(E/K_{ac})^{\Gamma_{ac}}$ is a pseudo-isomorphism.

Under some mild hypotheses, we show that $\text{Sel}_p(E/K_{ac}) \rightarrow \text{Sel}_p(E/K_\infty)^{H_{ac}}$ is an isomorphism and $H^1(H_{ac}, \text{Sel}_p(E/K_\infty)) = 0$.

Theorem 4.2. *Assume $p \nmid \prod_{v|p} \# \tilde{E}_v(k_v)$. Then*

- (1) *The restriction map*

$$\text{Sel}_p(E/K_{ac}) \rightarrow \text{Sel}_p(E/K_\infty)^{H_{ac}}$$

is an isomorphism,

- (2) $H^1(H_{ac}, \text{Sel}_p(E/K_\infty)) = 0$.

Proof. Consider the following commutative diagram:

$$\begin{array}{ccccccc}
0 & \longrightarrow & \mathrm{Sel}_p(E/K_\infty)^{H_{\mathrm{ac}}} & \longrightarrow & H^1(K_S/K_\infty, E_{p^\infty})^{H_{\mathrm{ac}}} & \xrightarrow{\lambda_\infty^{H_{\mathrm{ac}}}} & \bigoplus_{v \in S} J_v^1(E/K_\infty)^{H_{\mathrm{ac}}} \\
& & \uparrow \alpha & & \uparrow \beta & & \uparrow \gamma \\
0 & \longrightarrow & \mathrm{Sel}_p(E/K_{\mathrm{ac}}) & \longrightarrow & H^1(K_S/K_{\mathrm{ac}}, E_{p^\infty}) & \xrightarrow{\lambda_{\mathrm{ac}}} & \bigoplus_{v \in S} J_v^1(E/K_{\mathrm{ac}}) \longrightarrow 0.
\end{array}$$

The map λ_{ac} in this diagram is surjective because $\mathrm{Sel}_p(E/K_{\mathrm{ac}})$ is $\Lambda(\Gamma_{\mathrm{ac}})$ -cotorsion [HV03, Theorem 7.2]. We show that β is an isomorphism and $\gamma : \bigoplus_{v \in S} J_v^1(E/K_{\mathrm{ac}}) \rightarrow \bigoplus_{v \in S} J_v^1(E/K_\infty)^{H_{\mathrm{ac}}}$ is injective. It will then follow from the snake lemma that α is an isomorphism.

First, $\ker(\beta) = H^1(K_\infty/K_{\mathrm{ac}}, E_{p^\infty}(K_\infty))$ by the Hochschild-Serre spectral sequence. It follows from (surj) in Hypothesis 4.1 that $E_{p^\infty}(K_\infty) = 0$ and thus $H^1(H_{\mathrm{ac}}, E_{p^\infty}(K_\infty)) = 0$.

Now, let us compute $\ker(\gamma) = \bigoplus_{v \in S} \gamma_v$. If $v \nmid p$, then for any primes η in K_{ac} and w in K_∞ lying above p , the restriction map

$$H^1(K_{\mathrm{ac}, \eta}, E_{p^\infty}) \rightarrow H^1(K_{\infty, w}, E_{p^\infty})$$

is the identity map because both $K_{\mathrm{ac}, \eta}$ and $K_{\infty, w}$ are the unique pro- p unramified extension of K_v . We consider the case $v \mid p$. Note that every prime $\eta \mid v$ in K_{ac} is ramified over K . Using Proposition 4.8 of [CG96], we have

$$\frac{H^1(K_{\infty, w}, E_{p^\infty})}{\mathrm{Im}(\kappa_{\infty, w})} = H^1(K_{\infty, w}, (\tilde{E}_v)_{p^\infty}) \text{ and } \frac{H^1(K_{\mathrm{ac}, \eta}, E_{p^\infty})}{\mathrm{Im}(\kappa_{\mathrm{ac}, \eta})} = H^1(K_{\mathrm{ac}, \eta}, (\tilde{E}_v)_{p^\infty})$$

for every prime η in K_{ac} above p and every prime w in K_∞ above η . For the second identity, we used the assumption that v is ramified in K_{ac} . The restriction map is simply

$$\gamma_w : H^1(K_{\mathrm{ac}, \eta}, (\tilde{E}_v)_{p^\infty}) \rightarrow H^1(K_{\infty, w}, (\tilde{E}_w)_{p^\infty}),$$

whose kernel is $H^1(K_{\infty, w}/K_{\mathrm{ac}, \eta}, (\tilde{E}_v)_{p^\infty}(k_{\infty, w}))$ where $k_{\infty, w}$ is the residual field of $K_{\infty, w}$. We have $H^1(K_{\infty, w}/K_{\mathrm{ac}, \eta}, (\tilde{E}_v)_{p^\infty}(k_{\infty, w})) = 0$ because $(\tilde{E}_v)_{p^\infty}(k_v) = 0$ under our assumption and $k_{\infty, w}/k_v$ is a cyclic pro- p extension.

In fact, γ is an isomorphism because $\mathrm{coker}(\gamma_v) = H^2(K_{\infty, w}/K_{\mathrm{ac}, \eta}, (\tilde{E}_v)_{p^\infty}(k_{\infty, w})) = 0$. As a result, $\lambda_\infty^{H_{\mathrm{ac}}}$ is surjective because λ_{ac} is surjective and both β, γ are isomorphisms.

Now, we show that $H^1(H_{\text{ac}}, \text{Sel}_p(E/K_\infty)) = 0$. Using the same argument as Lemma 2.6 of [HO10], the fact that $\text{Sel}_p(E/K_{\text{ac}})$ is $\Lambda(\Gamma_{\text{ac}})$ -cotorsion implies $\text{Sel}_p(E/K_\infty)$ is $\Lambda(G_\infty)$ -cotorsion.

By [HV03, Theorem 7.2], we have $H^2(K_S/K_\infty, E_{p^\infty}) = 0$ and exactness of the sequence

$$0 \rightarrow \text{Sel}_p(E/K_\infty) \rightarrow H^1(K_S/K_\infty, E_{p^\infty}) \rightarrow \bigoplus_{v \in S} J_v^1(E/K_\infty) \rightarrow 0. \quad (4.1)$$

By the same argument as [CSS03, Lemma 2.4], $H^2(K_S/K_\infty, E_{p^\infty}) = 0$ implies

$$H^1(H_{\text{ac}}, H^1(K_S/K_\infty, E_{p^\infty})) = 0.$$

Taking H_{ac} -cohomology of short exact sequence (4.1) gives the identity

$$\text{coker}(\lambda_\infty^{H_{\text{ac}}}) = H^1(H_{\text{ac}}, \text{Sel}_p(E/K_\infty)).$$

This group is trivial because it has been shown that $\lambda_\infty^{H_{\text{ac}}}$ is surjective. $\square$

We first recall the definition of a regular sequence and the depth of a module M over a ring R , following [Eis95].

Definition 4.3. Suppose R is a ring and let M be an R -module. A sequence of elements $f_1, \dots, f_k$ of R is called an M -regular sequence if the following conditions hold:

- (1) The element f_i is a non-zero divisor on $M/(f_1, \dots, f_{i-1})M$ for each $i = 1, \dots, k$;
- (2) the module $M/(f_1, \dots, f_k)M$ is not zero.

If I is an ideal of the ring R and $f_1, \dots, f_k \in I$ then we call $f_1, \dots, f_k$ an M -regular sequence in I . Moreover, if $M = R$, we call $f_1, \dots, f_k$ a regular sequence in I .

Definition 4.4. Let I be an ideal of a ring R and suppose M is a finitely generated R -module such that $IM \neq M$. Then the I -depth of M , denoted by $\text{depth}_I(M)$, is the maximal length of a M -regular sequence in I . When R is a local ring, the depth of M as an R -module, denoted by $\text{depth}(M)$, is defined to be

$$\text{depth}(M) := \text{depth}_{\mathfrak{m}}(M)$$

where $\mathfrak{m}$ is the maximal ideal of R .

Theorem 4.5. *Assume the same hypotheses as Theorem 4.2. Then the $\Lambda(G_\infty)$ -module $X(E/K_\infty)$ does not have any non-trivial pseudo-null $\Lambda(G_\infty)$ -submodules.*

Proof. We will show $\text{pd}_{\Lambda(G_\infty)}(X(E/K_\infty)) \leq 1$, which implies that $X(E/K_\infty)$ does not have any non-trivial pseudo-null $\Lambda(G_\infty)$ -submodules by Proposition 3.2. We denote by $\text{depth}(X(E/K_\infty))$ the depth of $X(E/K_\infty)$ as a $\Lambda(G_\infty)$ -module. By the Auslander-Buchsbaum formula [Eis95, Theorem 19.9], the condition $\text{pd}_{\Lambda(G_\infty)}(X(E/K_\infty)) \leq 1$ is equivalent to $\text{depth}(X(E/K_\infty)) \geq 2$. We will compute $\text{depth}(X(E/K_\infty))$ using regular sequences.

Let γ_{cyc} and γ_{ac} be topological generators of Γ_{cyc} and Γ_{ac} , respectively. One may write $G_\infty = \Gamma_{\text{ac}} \times H_{\text{ac}}$ where $H_{\text{ac}} \simeq \Gamma_{\text{cyc}}$, so that γ_{cyc} can be regarded as a topological generator of H_{ac} . Denote $T_{\text{cyc}} := \gamma_{\text{cyc}} - 1$ and $T_{\text{ac}} := \gamma_{\text{ac}} - 1$. We show that $\{T_{\text{cyc}}, T_{\text{ac}}\}$ is an $X(E/K_\infty)$ -regular sequence in the maximal ideal $\mathfrak{m}$ of $\Lambda(G_\infty)$.

By Theorem 3.4, $X(E/K_\infty)[T_{\text{cyc}}] = H_1(H_{\text{ac}}, X(E/K_\infty)) = 0$. Therefore, T_{cyc} is a non-zero divisor on $X(E/K_\infty)$. Moreover, $X(E/K_\infty)/T_{\text{cyc}}X(E/K_\infty)$ is non-zero because otherwise $X(E/K_\infty)$ would be trivial by Nakayama's lemma.

On the other hand, T_{ac} is a non-zero divisor on $X(E/K_\infty)/T_{\text{cyc}}X(E/K_\infty)$ because

$$X(E/K_\infty)/T_{\text{cyc}}X(E/K_\infty) \simeq X(E/K_{\text{ac}})$$

by Theorem 4.2 and we have seen that $X(E/K_{\text{ac}})[T_{\text{ac}}] = H_1(\Gamma_{\text{ac}}, X(E/K_{\text{ac}})) = 0$ (Theorem 3.4). Once again, Nakayama's lemma implies

$$X(E/K_\infty)/(T_{\text{cyc}}, T_{\text{ac}})X(E/K_\infty)$$

is non-trivial. □

4.2. The indefinite setting. Throughout this section, we consider the hypotheses below, which are part of Assumptions A and B in the work of Bertolini [Ber01]:

Hypothesis 4.6. Assume that the following hold:

(good) $p \nmid 6N \text{disc}(K) \# \text{Pic}(\mathcal{O}_K)$.

(ord) E has good ordinary reduction at p .

(Heeg) The square-free part of N^- is a product of an even number of primes.

Hypothesis [\(Heeg\)](#) is called the weak Heegner hypothesis. In this setting, there is an explicitly constructed Heegner point $y_K \in E(K)$. In the simplest case when $N^- = 1$ (the strong Heegner hypothesis), there is an ideal $\mathfrak{N} \subset \mathcal{O}_K$ such that $N\mathcal{O}_K = \mathfrak{N}\overline{\mathfrak{N}}$ and y_K is the image of $[(\mathbb{C}/\mathcal{O}_K, \mathfrak{N}^{-1}/\mathcal{O}_K)]$ under the modular parametrization $\varphi : X_0(N) \rightarrow E$ sending $i\infty$ to the origin of E . We will assume that

(inf) y_K has infinite order in $E(K)$.

Note that our hypotheses are easier to check than the literature. For example, the conditions in Corollary [4.18](#) are numerically verifiable (see Example [4.19](#) and Example [4.20](#)).

Remark 4.7. The works of Kolyvagin [[Kol90](#)], Kolyvagin-Logachev [[KL89](#)] and Howard [[How04b](#)] imply the following:

(M-W) E/K has Mordell-Weil rank equal to 1,

(Sha) $\text{III}(E/K)_{p^\infty}$ is finite.

Under the hypothesis $N^- = 1$, Perrin-Riou constructed the free $\Lambda(\Gamma_{\text{ac}})$ -module of Heegner points $\mathcal{H}_{\text{ac}} \subset \mathcal{S}_p(E/K_{\text{ac}})$ [[PR87](#)]. This construction was later generalized by Howard under the weak Heegner hypothesis [[How04b](#)]. It follows from the norm relations on Heegner points that $\mathcal{H}_{\text{ac}}$ is $\Lambda(\Gamma_{\text{ac}})$ -free of rank 1 with our assumption on y_K (see for example, [[How04b](#), Theorem 3.4.3]). Even without this assumption, Cornut and Vatsal [[Cor02](#), [Vat03](#)] showed that $\mathcal{H}_{\text{ac}}$ is always non-trivial. Moreover, Howard showed that both $\widehat{\text{Sel}}_p(E/K_{\text{ac}})$ and $\mathcal{S}_p(E/K_{\text{ac}})$ have $\Lambda(\Gamma_{\text{ac}})$ -rank 1 assuming [\(inf\)](#) [[How04a](#), [How04b](#)]. The latter is a free $\Lambda(\Gamma_{\text{ac}})$ -module because there is an injection $\mathcal{S}_p(E/K_{\text{ac}}) \hookrightarrow \text{Hom}_\Lambda(X(E/K_{\text{ac}}), \Lambda)$ [[PR87](#), Lemme 5].

The non-triviality of Heegner points also implies the weak Leopoldt conjecture, i.e. the vanishing of $H^2(K_S/K_{\text{ac}}, E_{p^\infty})$ under certain hypotheses [[Ber01](#), Theorem 5.4]. We begin with a simpler proof of this fact assuming [\(inf\)](#).

Lemma 4.8. *Under our assumptions as in Hypothesis [4.6](#), we have the following:*

- (1) $H^2(K_S/K, E_{p^\infty}) = 0$.
- (2) $H^1(\Gamma_{\text{ac}}, H^1(K_S/K_{\text{ac}}, E_{p^\infty})) = 0$.
- (3) $H^2(K_S/K_{\text{ac}}, E_{p^\infty}) = 0$.

Proof. Recall that $E(K)$ has $\mathbb{Z}$ -rank 1 and $\text{III}(E/K)(p)$ is finite (see Remark 4.7). The same proof as Theorem 12 of [CM94] allows us to conclude that $H^2(K_S/K, E_{p^\infty}) = 0$, which is the strong Leopoldt conjecture.

We look at the Hochschild-Serre spectral sequence

$$H^i(\Gamma_{\text{ac}}, H^j(K_S/K_{\text{ac}}, E_{p^\infty})) \Rightarrow H^{i+j}(K_S/K, E_{p^\infty}).$$

The fact that $H^2(K_S/K, E_{p^\infty}) = 0$ and $H^2(\Gamma_{\text{ac}}, H^0(K_S/K_{\text{ac}}, E_{p^\infty})) = 0$ (Γ_{ac} has p -cohomological dimension 1) implies both part (ii) and part (iii) of the Lemma. $\square$

Throughout the rest of this section, we will analyze the fundamental diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Sel}_p(E/K_{\text{ac}})^{\Gamma_{\text{ac}}} & \longrightarrow & H^1(K_S/K_{\text{ac}}, E_{p^\infty})^{\Gamma_{\text{ac}}} & \xrightarrow{\lambda_{\text{ac}}^{\Gamma_{\text{ac}}}} & \bigoplus_{v \in S} J_v^1(E/K_{\text{ac}})^{\Gamma_{\text{ac}}} \\ & & \alpha \uparrow & & \beta \uparrow & & \gamma \uparrow \\ 0 & \longrightarrow & \text{Sel}_p(E/K) & \longrightarrow & H^1(K_S/K, E_{p^\infty}) & \xrightarrow{\lambda} & \bigoplus_{v \in S} H^1(K_v, E)(p). \end{array} \quad (4.2)$$

One may split (4.2) as follows:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Sel}_p(E/K_{\text{ac}})^{\Gamma_{\text{ac}}} & \longrightarrow & H^1(K_S/K_{\text{ac}}, E_{p^\infty})^{\Gamma_{\text{ac}}} & \xrightarrow{\phi_{\text{ac}}} & \text{Im}(\lambda_{\text{ac}}^{\Gamma_{\text{ac}}}) \longrightarrow 0 \\ & & \alpha \uparrow & & \beta \uparrow & & \delta \uparrow \\ 0 & \longrightarrow & \text{Sel}_p(E/K) & \longrightarrow & H^1(K_S/K, E_{p^\infty}) & \xrightarrow{\lambda} & \text{Im}(\lambda) \longrightarrow 0, \end{array} \quad (4.3)$$

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{Im}(\lambda_{\text{ac}}^{\Gamma_{\text{ac}}}) & \longrightarrow & \bigoplus_{v \in S} J_v^1(E/K_{\text{ac}})^{\Gamma_{\text{ac}}} & \longrightarrow & \text{coker}(\lambda_{\text{ac}}^{\Gamma_{\text{ac}}}) \longrightarrow 0 \\ & & \delta \uparrow & & \gamma \uparrow & & \epsilon \uparrow \\ 0 & \longrightarrow & \text{Im}(\lambda) & \longrightarrow & \bigoplus_{v \in S} H^1(K_v, E)(p) & \longrightarrow & \text{coker}(\lambda) \longrightarrow 0. \end{array} \quad (4.4)$$

Let ϕ_{ac} denote the map $H^1(K_S/K_{\text{ac}}, E_{p^\infty})^{\Gamma_{\text{ac}}} \rightarrow \text{Im}(\lambda_{\text{ac}}^{\Gamma_{\text{ac}}})$ in the top row of (4.3), thereby distinguishing it from the map $\lambda_{\text{ac}}^{\Gamma_{\text{ac}}}$ in (4.2). For each prime v in K dividing N , let $c_v(E/K)$ be the local Tamagawa factor of E/K at v . Moreover, for each prime $v \mid p$, we let $d_v = \#\tilde{E}_v(k_v)$, where $\tilde{E}_v$ is the reduction of E/K at v and k_v is the residual field of K with respect to v .

Theorem 4.9. *The restriction map $\gamma : \bigoplus_{v \in S} H^1(K_v, E) \rightarrow \bigoplus_{v \in S} J_v^1(E/K_{ac})^{\Gamma_{ac}}$ is surjective with finite kernel of order $\prod_{v|N^+} c_v(E/K)^{(p)} \times \prod_{v|p} (d_v^{(p)})^2$.*

Proof. For each prime v in K , denote by $\gamma_v : H^1(K_v, E) \rightarrow \bigoplus_{v \in S} J_v^1(E/K_{ac})^{\Gamma_{ac}}$ the local component of γ at v .

We first consider a prime v not lying above p . If the rational prime below v is inert in K , then v is infinitely split in K_{ac}/K and γ_v is simply the identity map. Otherwise, v is finitely decomposed in K_{ac}/K and the same argument as [CS00, Lemma 3.4] shows that γ_v is surjective and $\ker(\gamma_v)$ is finite of order $c_v(E/K)^{(p)}$, where $c_v(E/K)^{(p)}$ is the largest p -power that divides $c_v(E/K)$. Note that $c_v(E/K)$ is trivial for primes $v \nmid N$.

Next, consider a prime v lying above p . Since every prime above p is ramified in K_{ac}/K , the same argument as [CS00, Proposition 3.5] shows that γ_v is surjective and $\ker(\gamma_v)$ is finite of order $(d_v^{(p)})^2$. The conclusion now follows. $\square$

Corollary 4.10. *Both $\ker(\alpha)$ and $\operatorname{coker}(\alpha)$ are finite, and $\#\operatorname{coker}(\alpha)$ divides $\prod_{v|N^+} c_v(E/K)^{(p)} \times \prod_{v|p} (d_v^{(p)})^2$.*

Proof. The proof follows directly from Theorem 4.9 by applying the snake lemma to (4.2) once we show that both $\ker(\beta)$ and $\operatorname{coker}(\beta)$ are finite.

By the Hochschild-Serre spectral sequence, $\ker(\beta) = H^1(\Gamma_{ac}, E_{p^\infty}(K_{ac}))$. This group is always finite, following the argument in [Gre99, Lemma 3.1]. Note that the argument does not assume the finiteness of $E_{p^\infty}(K_{ac})$. Moreover, $\operatorname{coker}(\beta) = H^2(\Gamma_{ac}, E_{p^\infty}(K_{ac})) = 0$ since $\Gamma_{ac} \simeq \mathbb{Z}_p$ has p -cohomological dimension 1. $\square$

Lemma 4.11. *Suppose that $E_{p^\infty}(K) = 0$. Then:*

- (1) *The map β is an isomorphism*
- (2) *The map α is injective.*
- (3) *There is an isomorphism $\ker(\delta) \simeq \operatorname{coker}(\alpha)$.*

Proof. Part (i) directly follows from the fact that

$$\ker(\beta) = H^1(\Gamma_{ac}, E_{p^\infty}(K_{ac})) = 0$$

under the assumption $E_{p^\infty}(K) = 0$. Part (ii) and (iii) follow from part (i) by chasing (4.6). $\square$

Lemma 4.12. *The kernel of ϵ is finite of order $\#\ker(\gamma)/\#\ker(\delta)$.*

Proof. We saw that $\ker(\gamma)$ is finite (Theorem 4.9), and β is always surjective (Corollary 4.10). Hence, it follows from (4.3) that δ is surjective. Moreover, (4.4) gives the following exact sequence:

$$0 \rightarrow \ker(\delta) \rightarrow \ker(\gamma) \rightarrow \ker(\epsilon) \rightarrow 0,$$

and the Lemma immediately follows. $\square$

However, we can say more about the size of $\ker(\delta)$. Every prime $\ell \mid N^+$ splits in $K/\mathbb{Q}$ as $\ell\mathcal{O}_K = v\bar{v}$, and so $c_v(E/K) = c_{\bar{v}}(E/K) = c_\ell(E/\mathbb{Q})$ where $c_\ell(E/\mathbb{Q})$ is the Tamagawa factor at ℓ of $E/\mathbb{Q}$. Thus $\#\ker(\gamma) = (\prod_{\ell \mid N^+} c_\ell(E/\mathbb{Q})^{(p)} \times \prod_{v \mid p} d_v^{(p)})^2$ is a square, and the following result shows that $\ker(\delta)$ is bounded by its squareroot.

Lemma 4.13. *If p is split in $\mathcal{O}_K$, then $\#\ker(\delta) \mid \prod_{\ell \mid N^+} c_\ell(E/\mathbb{Q})^{(p)} \times \prod_{v \mid p} d_v^{(p)}$. If p is inert in $\mathcal{O}_K$ and $d_v^{(p)} = 1$ for the unique prime $v \mid p$, then $\#\ker(\delta) \mid \prod_{\ell \mid N^+} c_\ell(E/\mathbb{Q})^{(p)}$.*

Proof. Note that $\ker(\delta) = \text{Im}(\lambda) \cap \ker(\gamma)$. For each prime $\ell \mid N^+$ that splits as $(\ell) = v\bar{v}$ in $\mathcal{O}_K$, we have $G_v = G_{\bar{v}}$ in $\text{Gal}(K_S/K)$ where G_v and $G_{\bar{v}}$ are respectively the decomposition groups of v and $\bar{v}$. Hence, $\text{Im}(\lambda) \cap (\ker(\gamma_v) \times \ker(\gamma_{\bar{v}}))$ lies in the diagonal of $H^1(\Gamma_v, E)(p) \times H^1(\Gamma_{\bar{v}}, E)(p)$. When p is split in K , a similar argument also holds for the primes above p , and the conclusion follows.

On the other hand, it is not clear if the conclusion holds generally when p is inert. However, under the simplifying assumption $d_v^{(p)} = 1$ for $v \mid p$, the same argument implies $\#\ker(\delta) \mid \prod_{\ell \mid N^+} c_\ell(E/\mathbb{Q})^{(p)}$. $\square$

Under the weak Leopoldt conjecture (proved in Lemma 4.8), there is an exact sequence

$$0 \rightarrow \text{Sel}_p(E/K_{ac}) \rightarrow H^1(K_S/K_{ac}, E_{p^\infty}) \xrightarrow{\lambda_{ac}} \bigoplus_{v \in S} J_v^1(E/K_{ac}) \rightarrow \mathcal{S}_p(\widehat{E/K_{ac}}) \rightarrow 0. \quad (4.5)$$

We now state and prove the main theorem of this section.

Theorem 4.14. *The homology group $H_1(\Gamma_{ac}, X(E/K_{ac}))$ is finite of order*

$$\frac{\#\ker(\delta) \cdot [\mathcal{S}_p(E/K) : \text{Im}(\mathcal{S}_p(E/K_{ac})_{\Gamma_{ac}})]}{\#\ker(\gamma)}.$$

Thus, $H_1(\Gamma_{ac}, X(E/K_{ac})) = 0$ if and only if

$$\# \ker(\gamma) = [\mathcal{S}_p(E/K) : \text{Im}(\mathcal{S}_p(E/K_{ac})_{\Gamma_{ac}})] \# \ker(\delta).$$

Proof. Recall the exact sequence 4.5, which we may split into two short exact sequences

$$0 \rightarrow \text{Sel}_p(E/K_{ac}) \rightarrow H^1(K_S/K_{ac}, E_{p^\infty}) \xrightarrow{\lambda_{ac}} \text{Im}(\lambda_{ac}) \rightarrow 0, \quad (4.6)$$

$$0 \rightarrow \text{Im}(\lambda_{ac}) \rightarrow \bigoplus_{v \in S} J_v^1(E/K_{ac}) \rightarrow \mathcal{S}_p(\widehat{E/K_{ac}}) \rightarrow 0. \quad (4.7)$$

Because $H^1(\Gamma_{ac}, H^1(K_S/K_{ac}, E_{p^\infty})) = 0$ (Lemma 4.8), taking Γ_{ac} -invariants of the short exact sequence (4.6) gives the following exact sequence:

$$\begin{aligned} 0 \rightarrow \text{Sel}_p(E/K_{ac})^{\Gamma_{ac}} \rightarrow H^1(K_S/K_{ac}, E_{p^\infty})^{\Gamma_{ac}} \rightarrow (\text{Im}(\lambda_{ac}))^{\Gamma_{ac}} \\ \rightarrow H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) \rightarrow 0. \end{aligned} \quad (4.8)$$

Hence, $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) \simeq (\text{Im}(\lambda_{ac}))^{\Gamma_{ac}} / \text{Im}(\phi_{ac})$. On the other hand, taking Γ_{ac} -invariants of the short exact sequence (4.7) gives exactness of

$$0 \rightarrow (\text{Im}(\lambda_{ac}))^{\Gamma_{ac}} \rightarrow \bigoplus_{v \in S} J_v^1(E/K_{ac})^{\Gamma_{ac}} \rightarrow \mathcal{S}_p(\widehat{E/K_{ac}})^{\Gamma_{ac}} \rightarrow H^1(\Gamma_{ac}, \text{Im}(\lambda_{ac})). \quad (4.9)$$

Now, (4.6) again implies that there is an exact sequence

$$H^1(\Gamma_{ac}, H^1(K_S/K_{ac}, E_{p^\infty})) \rightarrow H^1(\Gamma_{ac}, \text{Im}(\lambda_{ac})) \rightarrow H^2(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) \quad (4.10)$$

Note that $H^1(\Gamma_{ac}, H^1(K_S/K_{ac}, E_{p^\infty})) = 0$ (Lemma 4.8) and $H^2(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = 0$ because Γ_{ac} has p -cohomological dimension 1. Thus, $H^1(\Gamma_{ac}, \text{Im}(\lambda_{ac})) = 0$, which lets us rewrite (4.9) as

$$0 \rightarrow (\text{Im}(\lambda_{ac}))^{\Gamma_{ac}} \rightarrow \bigoplus_{v \in S} J_v^1(E/K_{ac})^{\Gamma_{ac}} \rightarrow \mathcal{S}_p(\widehat{E/K_{ac}})^{\Gamma_{ac}} \rightarrow 0. \quad (4.11)$$

It follows that

$$0 \rightarrow \frac{(\text{Im}(\lambda_{ac}))^{\Gamma_{ac}}}{\text{Im}(\phi_{ac})} \rightarrow \frac{\bigoplus_{v \in S} J_v^1(E/K_{ac})^{\Gamma_{ac}}}{\text{Im}(\phi_{ac})} \rightarrow \mathcal{S}_p(\widehat{E/K_{ac}})^{\Gamma_{ac}} \rightarrow 0 \quad (4.12)$$

is exact. The group $\bigoplus_{v \in S} J_v^1(E/K_{ac})^{\Gamma_{ac}} / \text{Im}(\phi_{ac})$ is in fact $\text{coker}(\lambda_{ac}^{\Gamma_{ac}})$ in (4.5), and therefore $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) \simeq (\text{Im}(\lambda_{ac}))^{\Gamma_{ac}} / \text{Im}(\phi_{ac})$ is the kernel of the surjection

$$\text{coker}(\lambda_{ac}^{\Gamma_{ac}}) \rightarrow \mathcal{S}_p(\widehat{E/K_{ac}})^{\Gamma_{ac}}$$

induced by (4.5). Consider the following commutative diagram:

$$\begin{array}{ccc}
 \text{coker}(\lambda_{\text{ac}}^{\Gamma_{\text{ac}}}) & \longrightarrow & \widehat{\mathcal{S}_p(E/K_{\text{ac}})}^{\Gamma_{\text{ac}}} \\
 \uparrow \epsilon & \nearrow & \\
 \text{coker}(\lambda) = \widehat{\mathcal{S}_p(E/K)} & &
 \end{array} \tag{4.13}$$

We saw in Lemma 4.12 that ϵ is surjective with kernel of finite order

$$\# \ker(\gamma) / \# \text{coker}(\alpha).$$

On the other hand, the kernel of the diagonal map is finite of order

$$[\mathcal{S}_p(E/K) : \text{Im}(\mathcal{S}_p(E/K_{\text{ac}})_{\Gamma_{\text{ac}}})].$$

Hence, the kernel of the horizontal map, which is $H^1(\Gamma_{\text{ac}}, \text{Sel}_p(E/K_{\text{ac}}))$, is also finite. Its order is

$$\frac{[\mathcal{S}_p(E/K) : \text{Im}(\mathcal{S}_p(E/K_{\text{ac}})_{\Gamma_{\text{ac}}})]}{\# \ker(\epsilon)} = \frac{[\mathcal{S}_p(E/K) : \text{Im}(\mathcal{S}_p(E/K_{\text{ac}})_{\Gamma_{\text{ac}}})] \# \ker(\delta)}{\# \ker(\gamma)},$$

as required. $\square$

Write $K_{\text{ac}} = \cup_{n \geq 0} K_n$ and let $U\mathcal{S}_p(E/K)$ be the module of universal norms

$$U\mathcal{S}_p(E/K) := \bigcap_{n \geq 0} \text{Cores}_{K_n/K} \mathcal{S}_p(E/K_n)$$

inside $\mathcal{S}_p(E/K)$. If $\text{Im}(\mathcal{S}_p(E/K_{\text{ac}})_{\Gamma_{\text{ac}}}) = \mathcal{S}_p(E/K)$, then the corestriction maps $\text{Cores}_{K_n/K}$ are surjective, which implies $U\mathcal{S}_p(E/K) = \mathcal{S}_p(E/K)$. The converse also holds, for if $\text{Cores}_{K_n/K}$ are surjective then the natural map

$$\varprojlim_n \mathcal{S}_p(E/K_n) \rightarrow \mathcal{S}_p(E/K)$$

is also surjective because each $\mathcal{S}_p(E/K_n)$ is a finite-length $\mathbb{Z}_p$ -module. With respect to studying finite submodules of $X(E/K_{\text{ac}})$, Bertolini proved

Theorem 4.15. [Ber01, Corollary 6.2] *Assume*

(1) $\bar{\rho}_{E,p} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_p)$ is surjective,

(2) $p \nmid \prod_{v|p} d_v \cdot \prod_{\ell|N} c_{\ell}(E/\mathbb{Q})$.

Then the module $X(E/K_{\text{ac}})$ does not have any non-trivial finite $\Lambda(\Gamma_{\text{ac}})$ -submodule if and only if $\mathcal{S}_p(E/K)/U\mathcal{S}_p(E/K)$ is torsion-free.

Section 7 in *loc. cit.* also gives some conditions under which $\mathcal{S}_p(E/K)/U\mathcal{S}_p(E/K)$ is torsion-free. In particular,

Theorem 4.16. [Ber01, Theorem 7.1] *Assume the same hypotheses as Theorem 4.15 and*

- (1) *The module of Heegner points over K_n is non-zero modulo $pE(K_n)$ for some $n \geq 0$,*
- (2) *The natural map $\text{III}(E/K_n) \rightarrow \text{III}(E/K_{n+1})$ is an injection for all $n \geq 0$.*

Then $\mathcal{S}_p(E/K)/U\mathcal{S}_p(E/K)$ is torsion-free, and $X(E/K_{ac})$ does not have any non-trivial finite $\Lambda(\Gamma_{ac})$ -submodule.

In our setting, when (inf) holds, both $\mathcal{S}_p(E/K)$ and $U\mathcal{S}_p(E/K)$ have $\mathbb{Z}_p$ -rank 1. Therefore, the criterion $\mathcal{S}_p(E/K)/U\mathcal{S}_p(E/K)$ being torsion-free in Theorem 4.15 is equivalent to $\mathcal{S}_p(E/K) = U\mathcal{S}_p(E/K)$. We will give another proof of Theorem 3.1 in our setting using Theorem 4.14. First, we recall that $\text{Sel}_p(E/K_{ac})$ has $\Lambda(\Gamma_{ac})$ -corank 1 which is also the $\mathbb{Z}_p$ -corank of $\text{Sel}_p(E/K)$. Recall from Proposition 3.3 that $X(E/K_{ac})$ does not have any finite $\Lambda(\Gamma_{ac})$ -submodule if $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = 0$. We show that the condition $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = 0$ is equivalent to $\mathcal{S}_p(E/K) = U\mathcal{S}_p(E/K)$ under the same hypotheses as [Ber01].

Theorem 4.17. *Suppose that $\bar{\rho}_{E,p} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_{p^\infty})$ is surjective and $p \nmid \prod_{v|p} d_v \cdot \prod_{\ell|N} c_v(E/\mathbb{Q})$. Then $X(E/K_{ac})$ does not have any non-trivial finite $\Lambda(\Gamma_{ac})$ -submodule if and only if $U\mathcal{S}_p(E/K) = \mathcal{S}_p(E/K)$.*

Proof. Under our assumptions, we may use Corollary 4.10 to conclude that $\ker(\gamma)$ and $\ker(\delta)$ are trivial. The identity in Theorem 4.14 then becomes $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = [\mathcal{S}_p(E/K) : \text{Im}(\mathcal{S}_p(E/K_{ac})_{\Gamma_{ac}})]$, and we have seen that this is trivial if and only if $U\mathcal{S}_p(E/K) = \mathcal{S}_p(E/K)$. $\square$

It is natural to ask if $X(E/K_{ac})$ has no non-trivial finite $\Lambda(\Gamma_{ac})$ -submodules even if one relaxes the hypotheses in Theorem 4.17. To this end, we impose the strong Heegner hypothesis $N^- = 1$ and recall that the Birch & Swinnerton-Dyer (BSD) conjecture is equivalent to the identity

$$[E(K) : \mathbb{Z}y_K] = (\#\text{III}(E/K))^{1/2} u_K c_{\text{Manin}}(\varphi) \prod_{\ell|N} c_\ell(E/\mathbb{Q})$$

where $c_{\text{Manin}}(\varphi) \in \mathbb{Z}_{>0}$ is the Manin constant for φ and $u_K = \#(\mathcal{O}_K^\times/\mathbb{Z}^\times) = 1$. It is worth noting that $u_K = 1$ under our assumption on K and that $c_{\text{Manin}}(\varphi)$ is conjectured to be 1. We refer the readers to [GZ86, (2.2)] for this formulation of the BSD conjecture and [ARS06] for the conjecture regarding the Manin constant.

Granting these conjectures, p does not divide $\text{III}(E/K)$ if and only if

$$[E(K) \otimes \mathbb{Z}_p : \mathbb{Z}_p(y_K \otimes 1)] = \prod_{\ell|N} c_\ell(E/\mathbb{Q})^{(p)}.$$

Under this assumption, we extend the result of Bertolini [Ber01] to cases where p divides the product $\prod_{\ell|N} c_\ell(E/\mathbb{Q}) \cdot \prod_{v|p} d_v$, or when $\bar{\rho}_{E,p} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_p)$ is not surjective.

Corollary 4.18. *Assume $N^- = 1$. Suppose $[E(K) \otimes \mathbb{Z}_p : \mathbb{Z}_p(y_K \otimes 1)] = \prod_{\ell|N} c_\ell(E/\mathbb{Q})^{(p)}$ and that one of the following holds:*

- (1) *The prime p is split in $\mathcal{O}_K$.*
- (2) *The prime p is inert in $\mathcal{O}_K$ and $d_v^{(p)} = 1$ for the unique prime $v \mid p$ in K .*

Then $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = 0$ and the module $X(E/K_{ac})$ does not have any non-zero finite submodules.

Proof. Since $\text{III}(E/K)(p)$ is finite (see Remark 4.7), we may identify $E(K) \otimes \mathbb{Z}_p$ with $\mathcal{S}_p(E/K)$. Note that the image of $\mathcal{S}_p(E/K_{ac})_{\Gamma_{ac}}$ in $\mathcal{S}_p(E/K) = E(K) \otimes \mathbb{Z}_p$ contains $\mathbb{Z}_p \left(\prod_{v|p} d_v^{(p)} \right) (y_K \otimes 1)$ [PR95, 438]. Therefore $[\mathcal{S}_p(E/K) : \mathcal{S}_p(E/K_{ac})_{\Gamma_{ac}}]$ divides

$$\prod_{v|p} d_v^{(p)} \cdot [E(K) \otimes \mathbb{Z}_p : \mathbb{Z}_p(y_K \otimes 1)] = \prod_{\ell|N} c_\ell(E/\mathbb{Q})^{(p)} \cdot \prod_{v|p} d_v^{(p)}.$$

Once again, we note that this is precisely $(\# \ker(\gamma))^{1/2}$. Under our assumptions, $\# \ker(\delta)$ also divides the RHS by Lemma 4.13. It follows from Theorem 4.14 that $H^1(\Gamma_{ac}, \text{Sel}_p(E/K_{ac})) = 0$ and the module $X(E/K_{ac})$ does not have any non-zero finite submodules. $\square$

We illustrate our results with numerical examples which fall under the hypotheses of Corollary 4.18 where Theorem 4.15 does not apply.

Example 4.19. Consider the elliptic curve $E/\mathbb{Q}$ with Cremona label 11a1 and let $K = \mathbb{Q}(\sqrt{-19})$, $p = 5$. In this example, 5 is split in $\mathcal{O}_K$ and $[E(K) \otimes \mathbb{Z}_5 : \mathbb{Z}_5(y_K \otimes 1)] = c_{11}^{(5)} = 5$. The Heegner point y_K has

trivial index in the free part of $E(K)$ and $[E(K) \otimes \mathbb{Z}_5 : \mathbb{Z}_5(y_K \otimes 1)]$ is contributed by the 5-torsion points $E_5(K)$. Other important properties of this example are the residual representation $\bar{\rho}_{E,5} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_5)$ is not surjective and $d_v^{(5)} = 5$ for each prime $v \mid 5$.

Example 4.20. Consider the elliptic curve $E/\mathbb{Q}$ with Cremona label 326b1 and let $K = \mathbb{Q}(\sqrt{-31})$, $p = 5$. In this example, 5 is also split in $\mathcal{O}_K$ and $[E(K) \otimes \mathbb{Z}_5 : \mathbb{Z}_5(y_K \otimes 1)] = \prod_{\ell \mid N} c_{\ell}^{(5)} = 5$. In contrast with the previous example, $E(K)$ has trivial torsion and the Heegner point y_K has index 5 in $E(K)$. The residual representation $\bar{\rho}_{E,5} : G_{\mathbb{Q}} \rightarrow \text{Aut}(E_5)$ is surjective, and $d_v^{(5)} = 1$ for each prime $v \mid 5$.

We end this section by remarking that in the case where $p \nmid \prod_{v \mid p} d_v \cdot \prod_{v \mid N} c_v(E/K)$ and $y(K) \notin pE(K)$, Matar and Nekovář recently showed that $\text{Sel}_p(E/K_{\text{ac}})$ is in fact cofree if we additionally assume that E_p is an irreducible $G_{\mathbb{Q}}$ -module [MNr19].

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